



## **Semester I Examinations 2018/2019**

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|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exam Codes</b>              | 3BA1, 3BS9, 3MR3, 3BPT1, 4BA4, 4BME1, 4BS2, 1CNS1<br>1OA1, 1EM1                                                                         |
| <b>Exams</b>                   | Third Arts, Third Science, Third Marine Science,<br>Fourth Arts, Fourth Mathematics and Education, Fourth Science,<br>Master of Science |
| <b>Module</b>                  | Applied Statistics I                                                                                                                    |
| <b>Module Codes</b>            | <b>ST311</b>                                                                                                                            |
| External Examiner              | Professor S. Wilson                                                                                                                     |
| Internal Examiner(s)           | Dr. E. Holian<br>Professor J. Newell                                                                                                    |
| <b><u>Instructions:</u></b>    | <b>Answer any 3 questions from Questions 1-4</b><br><b>All Questions carry equal marks.</b>                                             |
| <b>Duration</b>                | 2 Hours                                                                                                                                 |
| <b>No. of Pages</b>            | 20 pages, including this one                                                                                                            |
| <b>School</b>                  | School of Mathematics, Statistics and Applied Mathematics                                                                               |
| Release to Library:            | Yes <input checked="" type="checkbox"/> No <input type="checkbox"/>                                                                     |
| Release in Exam Venue          | Yes <input checked="" type="checkbox"/> No <input type="checkbox"/>                                                                     |
| <b><u>Requirements:</u></b>    |                                                                                                                                         |
| Statistical Tables/ Log Tables | Relevant distribution tables are attached to this paper.<br>The <i>New Formulae &amp; Tables</i> are optional.                          |
| Other Materials Allowed        | A calculator is allowed (non-programmable and<br>not capable of storing text).                                                          |

## Question 1

Complete parts (a), (b) and (c).

- (a) McDundee take-away is responding to a customer complaint regarding the weight of medium sized portions of fries. A sample of medium portion french-fries are taken at random and the weights (in grams) recorded:

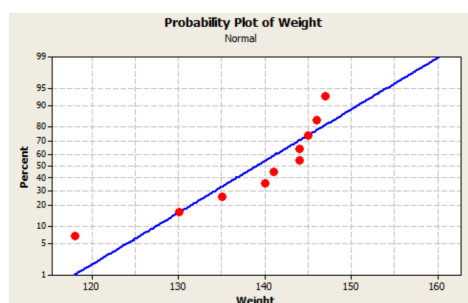
118, 130, 135, 140, 141, 144, 144, 145, 146, 147.

A summary of descriptive statistics and plots for this sample are provided below:

| Variable | N  | N* | Mean   | SE Mean | StDev | Minimum | Q1     | Median | Q3     |
|----------|----|----|--------|---------|-------|---------|--------|--------|--------|
| Weight   | 10 | 0  | 139.00 | 2.87    | 9.08  | 118.00  | 133.75 | 142.50 | 145.25 |

| Variable | Maximum | Mode | N for Mode | Skewness | Kurtosis |
|----------|---------|------|------------|----------|----------|
| Weight   | 147.00  | 144  | 2          | -1.62    | 2.42     |



- By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]
  - What does the normal probability plot suggest? [2 marks]
  - Construct the letter value display for this sample, including mid-values and spreads. [4 marks]
  - What do the letter values indicate about the sample distribution? [4 marks]
- (b) The take-away manager wishes to determine whether this sample gives evidence that the true typical weight of medium-portion french-fries is less than 145g.
- It is suggested that a 1-sample t-test be carried out. The output is included below. Assuming the assumptions of the test hold what can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

One-Sample T: Weight

Test of  $\mu = 145$  vs  $< 145$

95% Upper

| Variable | N  | Mean   | StDev | SE Mean | Bound  | T     | P     |
|----------|----|--------|-------|---------|--------|-------|-------|
| Weight   | 10 | 139.00 | 9.08  | 2.87    | 144.26 | -2.09 | 0.033 |

Question 1 continues on the next page

Question 1 continued...

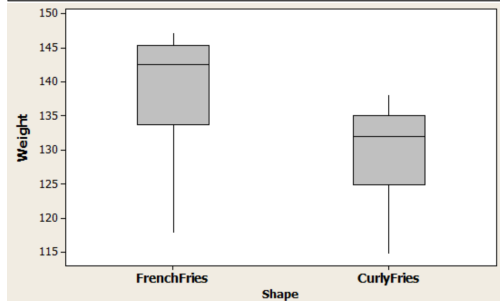
- (ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]
- (iii) It is suggested that a 1-sample Sign test be carried out. Use the sample to test where there is evidence to suggest  $\eta < 145$ , testing at  $\alpha = 0.05$ . Include a statement of the hypotheses being tested, an explanation of how the p-value is calculated defining any symbols you use, your decision and conclusion in this application. [7 marks]
- (c) The complaint received suggested that the typical weight of medium portion fries is different if the fries are a different shape, i.e straight FrenchFries compared to CurlyFries. To investigate, independent samples of each shape are taken and the weights recorded:

FrenchFries 118, 130, 135, 140, 141, 144, 144, 145, 146, 147.

CurlyFries 115, 125, 132, 135, 138, 126, 134.

Some descriptive statistics and plots for these samples are also provided.

| Variable    | N  | Mean   | SE Mean | StDev | Minimum | Q1     | Median | Q3     | Maximum |
|-------------|----|--------|---------|-------|---------|--------|--------|--------|---------|
| FrenchFries | 10 | 139.00 | 2.87    | 9.08  | 118.00  | 133.75 | 142.50 | 145.25 | 147.00  |
| CurlyFries  | 7  | 129.29 | 2.97    | 7.87  | 115.00  | 125.00 | 132.00 | 135.00 | 138.00  |



The manager wishes to carry out a hypothesis test to infer whether the samples give evidence that the typical weight for the population of FrenchFries is greater than the typical value of weight for the population of CurlyFries.

- (i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]
- (ii) Carry out the test at significance level  $\alpha=0.05$ . Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

## Question 2

Complete parts (a), (b) and (c).

- (a) (i) "The true correlation coefficient between two variables  $X$  and  $Y$  for a population of individuals is known to have value  $\rho = 0$ . This implies that  $X$  and  $Y$  are unrelated."
- Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]
- (ii) A sample of 15 individuals gave a sample correlation coefficient of  $r = 0.59$ . Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between  $X$  and  $Y$  in the population. Include clear statements of the hypotheses, the rejection region and conclusion. [6 marks]
- (b) Consider a simple linear regression model  $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ .
- (i) What does the error term,  $\epsilon_i$ , represent? [2 marks]
- (ii) When carrying out an Analysis of Variance, explain in general what the terms  $SS_{Total}$ ,  $SS_{Reg}$  and  $SS_{Error}$  represent. [5 marks]
- (c) The following model is being considered in order to model the variability in a response variable  $Y$ , using five candidate predictor variables,  $X_1, X_2, X_3, X_4, X_5$ ,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i.$$

The model is fitted to observations obtained for a sample of 15 individuals, and the results of the fit are provided in the partial printout on the next page.

Use the output to answer the following questions.

- (i) Calculate the coefficient of determination  $R^2$  for this regression fit. Interpret the meaning of this value. [2 marks]
- (ii) Find the missing values in the ANOVA table, and complete the test, at the  $\alpha = 0.05$  level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]
- (iii) Complete the missing t-test in the output, i.e. test whether  $X_2$  is a good predictor of  $Y$  at the  $\alpha = 0.05$  level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]
- (iv) Use the partial F-test to test the hypothesis :  $H_0 : \beta_4 = \beta_5 = 0$  at  $\alpha = 0.05$ . Interpret your findings. [5 marks]
- (v) The output provides a prediction interval and a confidence interval for the response when  $X_1 = 7$ ,  $X_2 = 40$ ,  $X_3 = 13$ ,  $X_4 = 25$ ,  $X_5 = 1$ . Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

The regression equation is

$$Y = 19.72 + 2.38 X_1 + 0.37 X_2 - 0.44 X_3 + 0.82 X_4 + 6.08 X_5$$

| Predictor | Coef  | SE Coef | T     | P     |
|-----------|-------|---------|-------|-------|
| Constant  | 19.72 | 17.14   | 1.15  | 0.280 |
| X1        | 2.38  | 0.684   | 3.48  | 0.007 |
| X2        | 0.37  | 0.090   | -     | -     |
| X3        | -0.44 | 0.188   | -2.34 | 0.045 |
| X4        | 0.82  | 0.511   | 1.60  | 0.143 |
| X5        | 6.08  | 4.686   | 1.30  | 0.226 |

S = 8.851      R-Sq = - %      R-Sq(adj) = - %

Analysis of Variance

| Source         | DF | SS   | MS | F | P |
|----------------|----|------|----|---|---|
| Regression     | -  | 2313 | -  | - | - |
| Residual Error | -  | -    | -  | - | - |
| Total          | -  | 3018 |    |   |   |

| Source | DF | Seq SS |
|--------|----|--------|
| X1     | 1  | 253    |
| X2     | 1  | 1301   |
| X3     | 1  | 425    |
| X4     | 1  | 202    |
| X5     | 1  | 132    |

Predicted Values for New Observations

| NewObs | Fit   | SE Fit | 95% CI         | 95% PI         |
|--------|-------|--------|----------------|----------------|
| 1      | 72.04 | 3.15   | (64.92, 79.16) | (50.79, 93.29) |

Values of Predictors for New Observations

| NewObs | X1   | X2   | X3   | X4   | X5   |
|--------|------|------|------|------|------|
| 1      | 7.00 | 40.0 | 13.0 | 25.0 | 1.00 |

[Total 35 marks]

### Question 3

Complete parts (a), (b), and (c).

Researchers running the Allen Cognitive Level (ACL) Study were interested in the relationship of ACL test scores to the level of psychopathology. The following candidate predictor variables were recorded for a sample of 23 patients:

- SDMT: score on the Symbol-Digit Modalities Test
- Vocab: vocabulary score on the Shipley Institute of Living Scale
- Abstract: abstraction score on the Shipley Institute of Living Scale

(a) The output below is produced by fitting a model where SDMT is used to predict ACL,  $ACL_i = \beta_0 + \beta_1 SDMT_i + \epsilon_i$ .

The regression equation is

ACL = 4.01 + 0.0252 SDMT

| Predictor | Coef   | SE Coef | T     | P     |
|-----------|--------|---------|-------|-------|
| Constant  | 4.01   | 0.3866  | 10.37 | 0.000 |
| SDMT      | 0.0252 | 0.00924 | 2.72  | 0.013 |

S = 0.7    R-Sq = 26.1%    R-Sq(adj) = 22.6%

Unusual Observations

| Obs | SDMT | ACL | Fit  | SE Fit | Residual | St Resid |
|-----|------|-----|------|--------|----------|----------|
| 16  | 61.0 | 3.9 | 5.54 | 0.252  | -1.64    | -2.51 R  |
| 18  | 2.0  | 3.4 | 4.06 | 0.370  | -0.66    | -1.11 X  |
| 23  | 47.0 | 6.6 | -    | 0.165  | -        | 2.01 R   |

R denotes an observation with a large standardized residual.

X denotes an observation whose X value gives it large leverage.

(i) Interpret the meaning of the fitted constant and coefficient in this model.

[4 marks]

(ii) The output references individuals in the sample that have a large standardised residual.

The last observation, with predictor value  $SDMT = 47$  and observed response  $ACL = 6.6$ , has some output values missing.

Find the value of the fitted response  $\hat{ACL}$  and the fitted residual  $\hat{\epsilon}$  for this individual, and thus show how the standardised residual of 2.01 is arrived at.

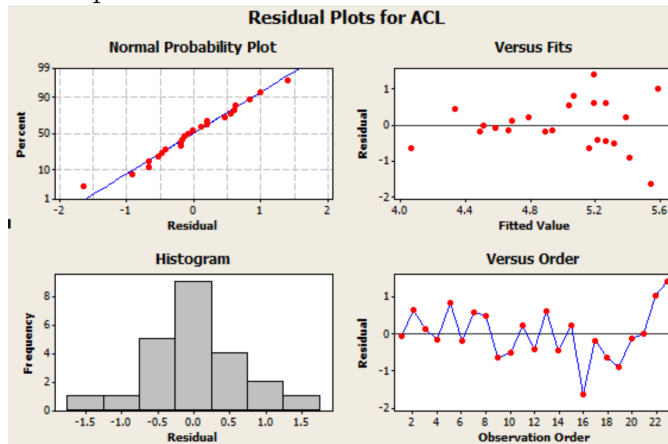
Why have these observations been flagged as unusual observations?

[5 marks]

Question 3 continues on the next page

Question 3 continued...

- (iii) Explain how the diagnostics plots are used to assess validity of the regression assumptions.



[4 marks]

- (b) The output below is produced by fitting a model where SDMT and Abstract are used to predict ACL,  $ACL_i = \beta_0 + \beta_1 SDMT_i + \beta_2 Abstract_i + \epsilon_i$ . Discuss the predictive power of the two variables, *SDMT* and *Abstract*, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

Regression Analysis: ACL versus SDMT, Abstract

The regression equation is

ACL = 3.98 + 0.0231 SDMT + 0.0045 Abstract

| Predictor | Coef    | SE Coef | T    | P     |
|-----------|---------|---------|------|-------|
| Constant  | 3.9836  | 0.4075  | 9.78 | 0.000 |
| SDMT      | 0.02314 | 0.01240 | 1.87 | 0.077 |
| Abstract  | 0.00453 | 0.01809 | 0.25 | 0.805 |

S = 0.716455    R-Sq = 26.3%    R-Sq(adj) = 18.9%

Analysis of Variance

| Source         | DF | SS      | MS     | F    | P     |
|----------------|----|---------|--------|------|-------|
| Regression     | 2  | 3.6669  | 1.8334 | 3.57 | 0.047 |
| Residual Error | 20 | 10.2662 | 0.5133 |      |       |
| Total          | 22 | 13.9330 |        |      |       |

| Source   | DF | Seq SS |
|----------|----|--------|
| SDMT     | 1  | 3.6346 |
| Abstract | 1  | 0.0322 |

[12 marks]

Question 3 continues on the next page

(c) A best-subsets routine was carried out specifying the 3 variables listed above as candidate predictors.

|      |      |           |     |         |     |       |
|------|------|-----------|-----|---------|-----|-------|
|      |      |           |     |         |     | A     |
|      |      |           |     |         |     | b     |
|      |      |           |     |         |     | s     |
|      |      |           |     |         |     | V t   |
|      |      |           |     |         |     | S o r |
|      |      |           |     |         |     | D c a |
|      |      |           |     |         |     | M a c |
|      |      |           |     |         |     | T b t |
| Vars | R-Sq | R-Sq(adj) | Cp  | S       |     |       |
| 1    | 26.1 | 22.6      | 0.1 | 0.70029 | X   |       |
| 1    | 13.5 | 9.4       | 3.3 | 0.75759 |     | X     |
| 2    | 26.4 | 19.0      | 2.0 | 0.71616 | X X |       |
| 2    | 26.3 | 18.9      | 2.0 | 0.71646 | X   | X     |
| 3    | 26.5 | 14.8      | 4.0 | 0.73440 | X X | X     |

- [Total 35 marks]



## Question 4

Complete parts (a), (b) and (c).

- (a) A researcher compared the petal length (PL) of three different Iris plant species.

Indicator variables are defined for the three species:

$$\begin{aligned} \text{setosa} : & \begin{cases} 1 & \text{if the plant observed is species 1, setosa} \\ 0 & \text{otherwise} \end{cases} \\ \text{vers} : & \begin{cases} 1 & \text{if the plant observed is species 2, a versicolor} \\ 0 & \text{otherwise} \end{cases} \\ \text{vir} : & \begin{cases} 1 & \text{if the plant observed is species 3, a virginica} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and the model,  $PL_i = \beta_0 + \beta_1 \text{setosa}_i + \beta_2 \text{vers}_i + \epsilon_i$ , is fitted to the data, giving:

The regression equation is

$$PL = 5.55 - 4.09 \text{ setosa} - 1.29 \text{ vers}$$

| Predictor | Coef  | SE Coef | T      | P     |
|-----------|-------|---------|--------|-------|
| Constant  | 5.55  | 0.06086 | 91.23  | 0.000 |
| setosa    | -4.09 | 0.08607 | -47.52 | 0.000 |
| vers      | -1.29 | 0.08607 | -15.01 | 0.000 |

S = 0.430334    R-Sq = 94.1%    R-Sq(adj) = 94.1%

- i) Use the fitted model to predict PL for:
  - a plant of species setosa.
  - a plant of species versicolor.
  - a plant of species virginica.

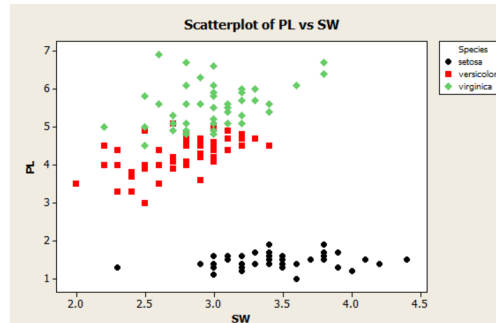
[3 marks]
- ii) Interpret the meaning of the estimated coefficients of *setosa* and *versicolor*, and comment on their significance. Ensure you state the hypotheses being tested when referring to the results of the output.

[6 marks]
- iii) Write down an alternative model proposal that you would recommend be fitted to the observed data that would help assess the comparison of PL between each of the species, versicolor and virginica, compared with the setosa species.

[2 marks]

Question 4 continues on the next page

- Question 4 continued...*
- (b) In an attempt to control for substantial variability in petal length, the sepal width (SW) of each plant was measured as a covariate. A plot of the observed relationship of PL with SW is provided, with an indication to the different species.



The regression analysis below models the response,  $PL$ , by the predictor  $SW$ , and by the indicator variables  $setosa$ ,  $vers$ , as defined above.

The regression equation is  
 $PL = 4.13 + 0.479 SW - 4.31 setosa - 1.19 vers$

| Predictor | Coef     | SE Coef | T      | P     |
|-----------|----------|---------|--------|-------|
| Constant  | 4.1282   | 0.2942  | 14.03  | 0.000 |
| SW        | 0.47876  | 0.09707 | 4.93   | 0.000 |
| setosa    | -4.30736 | 0.09130 | -47.18 | 0.000 |
| vers      | -1.19433 | 0.08237 | -14.50 | 0.000 |

- Interpret the meaning of the fitted coefficients of  $SW$ ,  $setosa$  and  $vers$ , in this model. [5 marks]
  - For this model, write down the estimated relationship between  $PL$  and  $SW$  for each of the three species,  $setosa$ ,  $versicolor$  and  $virginica$ . [3 marks]
  - What assumption is being made about the relationship of  $PL$  and  $SW$  for the three different species in this model? [2 marks]
- (c) Two interaction terms were added to the model:  
 $SW.setosa$ , calculated as  $(SW_i) \times (setosa_i)$  for each individual,  
 $SW.vers$ , calculated as  $(SW_i) \times (vers_i)$  for each individual,  
giving the following output:

**Question 4 continues on the next page**

Question 4 continued...

The regression equation is  
 $PL = 3.51 + 0.686 SW - 2.33 \text{ setosa} - 1.58 \text{ vers} - 0.605 SW.\text{setosa} + 0.153 SW.\text{vers}$

| Predictor | Coef    | SE Coef | T     | P     |
|-----------|---------|---------|-------|-------|
| Constant  | 3.5109  | 0.5105  | 6.88  | 0.000 |
| SW        | 0.6863  | 0.1707  | 4.02  | 0.000 |
| setosa    | -2.3280 | 0.7151  | -3.26 | 0.001 |
| vers      | -1.5760 | 0.7068  | -2.23 | 0.027 |
| SW.setosa | -0.6049 | 0.2241  | -2.70 | 0.008 |
| SW.vers   | 0.1531  | 0.2447  | 0.63  | 0.533 |

S = 0.385285    R-Sq = 95.4%    R-Sq(adj) = 95.2%

- For this model, write down the estimated relationship between between PL and SW for each of the three species. [4 marks]
- Interpret the meaning of the fitted coefficients of *SW.setosa* and *SW.vers* in this model. [5 marks]
- Determine if the terms *SW.setosa* and *SW.vers* are significant, using the results of the individual t-tests in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output. [5 marks]

[Total 35 marks]

# ST311 Formulae

Some useful z-scores:

$$Z_{0.1} = 1.28$$

$$Z_{0.05} = 1.64$$

$$Z_{0.025} = 1.96$$

$$Z_{0.01} = 2.33$$

$$Z_{0.005} = 2.58$$

Some ranges for the standard normal distribution:

$$1.35 \quad \text{middle 50\% : } Z_{0.75} - Z_{0.25}$$

$$2.30 \quad \text{middle 75\% : } Z_{0.875} - Z_{0.125}$$

$$3.07 \quad \text{middle 87.5\% : } Z_{0.9375} - Z_{0.0625}$$

$$3.73 \quad \text{middle 93.75\% : } Z_{0.96875} - Z_{0.03125}$$

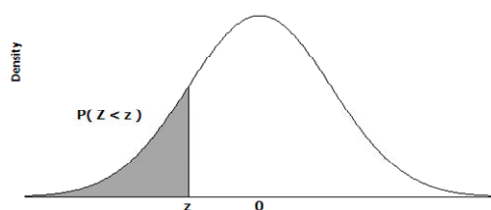
$$4.31 \quad \text{middle 96.875\% : } Z_{0.984375} - Z_{0.015625}$$

$$4.84 \quad \text{middle 98.4375\% : } Z_{0.9921875} - Z_{0.0078125}$$

Some formulae that may be useful when carrying out inference with parametric procedures:

- Inference for population correlation coefficient  $\rho$ :  $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test :  $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution,  $Z \sim \text{Normal}(0,1)$ .



| <b>Z</b>    | <b>0.00</b> | <b>0.01</b> | <b>0.02</b> | <b>0.03</b> | <b>0.04</b> | <b>0.05</b> | <b>0.06</b> | <b>0.07</b> | <b>0.08</b> | <b>0.09</b> |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>-3.4</b> | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0002      |
| <b>-3.3</b> | 0.0005      | 0.0005      | 0.0005      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0003      |
| <b>-3.2</b> | 0.0007      | 0.0007      | 0.0006      | 0.0006      | 0.0006      | 0.0006      | 0.0006      | 0.0005      | 0.0005      | 0.0005      |
| <b>-3.1</b> | 0.0010      | 0.0009      | 0.0009      | 0.0009      | 0.0008      | 0.0008      | 0.0008      | 0.0008      | 0.0007      | 0.0007      |
| <b>-3.0</b> | 0.0013      | 0.0013      | 0.0013      | 0.0012      | 0.0012      | 0.0011      | 0.0011      | 0.0011      | 0.0010      | 0.0010      |
| <b>-2.9</b> | 0.0019      | 0.0018      | 0.0018      | 0.0017      | 0.0016      | 0.0016      | 0.0015      | 0.0015      | 0.0014      | 0.0014      |
| <b>-2.8</b> | 0.0026      | 0.0025      | 0.0024      | 0.0023      | 0.0023      | 0.0022      | 0.0021      | 0.0021      | 0.0020      | 0.0019      |
| <b>-2.7</b> | 0.0035      | 0.0034      | 0.0033      | 0.0032      | 0.0031      | 0.0030      | 0.0029      | 0.0028      | 0.0027      | 0.0026      |
| <b>-2.6</b> | 0.0047      | 0.0045      | 0.0044      | 0.0043      | 0.0041      | 0.0040      | 0.0039      | 0.0038      | 0.0037      | 0.0036      |
| <b>-2.5</b> | 0.0062      | 0.0060      | 0.0059      | 0.0057      | 0.0055      | 0.0054      | 0.0052      | 0.0051      | 0.0049      | 0.0048      |
| <b>-2.4</b> | 0.0082      | 0.0080      | 0.0078      | 0.0075      | 0.0073      | 0.0071      | 0.0069      | 0.0068      | 0.0066      | 0.0064      |
| <b>-2.3</b> | 0.0107      | 0.0104      | 0.0102      | 0.0099      | 0.0096      | 0.0094      | 0.0091      | 0.0089      | 0.0087      | 0.0084      |
| <b>-2.2</b> | 0.0139      | 0.0136      | 0.0132      | 0.0129      | 0.0125      | 0.0122      | 0.0119      | 0.0116      | 0.0113      | 0.0110      |
| <b>-2.1</b> | 0.0179      | 0.0174      | 0.0170      | 0.0166      | 0.0162      | 0.0158      | 0.0154      | 0.0150      | 0.0146      | 0.0143      |
| <b>-2.0</b> | 0.0228      | 0.0222      | 0.0217      | 0.0212      | 0.0207      | 0.0202      | 0.0197      | 0.0192      | 0.0188      | 0.0183      |
| <b>-1.9</b> | 0.0287      | 0.0281      | 0.0274      | 0.0268      | 0.0262      | 0.0256      | 0.0250      | 0.0244      | 0.0239      | 0.0233      |
| <b>-1.8</b> | 0.0359      | 0.0351      | 0.0344      | 0.0336      | 0.0329      | 0.0322      | 0.0314      | 0.0307      | 0.0301      | 0.0294      |
| <b>-1.7</b> | 0.0446      | 0.0436      | 0.0427      | 0.0418      | 0.0409      | 0.0401      | 0.0392      | 0.0384      | 0.0375      | 0.0367      |
| <b>-1.6</b> | 0.0548      | 0.0537      | 0.0526      | 0.0516      | 0.0505      | 0.0495      | 0.0485      | 0.0475      | 0.0465      | 0.0455      |
| <b>-1.5</b> | 0.0668      | 0.0655      | 0.0643      | 0.0630      | 0.0618      | 0.0606      | 0.0594      | 0.0582      | 0.0571      | 0.0559      |
| <b>-1.4</b> | 0.0808      | 0.0793      | 0.0778      | 0.0764      | 0.0749      | 0.0735      | 0.0721      | 0.0708      | 0.0694      | 0.0681      |
| <b>-1.3</b> | 0.0968      | 0.0951      | 0.0934      | 0.0918      | 0.0901      | 0.0885      | 0.0869      | 0.0853      | 0.0838      | 0.0823      |
| <b>-1.2</b> | 0.1151      | 0.1131      | 0.1112      | 0.1093      | 0.1075      | 0.1056      | 0.1038      | 0.1020      | 0.1003      | 0.0985      |
| <b>-1.1</b> | 0.1357      | 0.1335      | 0.1314      | 0.1292      | 0.1271      | 0.1251      | 0.1230      | 0.1210      | 0.1190      | 0.1170      |
| <b>-1.0</b> | 0.1587      | 0.1562      | 0.1539      | 0.1515      | 0.1492      | 0.1469      | 0.1446      | 0.1423      | 0.1401      | 0.1379      |
| <b>-0.9</b> | 0.1841      | 0.1814      | 0.1788      | 0.1762      | 0.1736      | 0.1711      | 0.1685      | 0.1660      | 0.1635      | 0.1611      |
| <b>-0.8</b> | 0.2119      | 0.2090      | 0.2061      | 0.2033      | 0.2005      | 0.1977      | 0.1949      | 0.1922      | 0.1894      | 0.1867      |
| <b>-0.7</b> | 0.2420      | 0.2389      | 0.2358      | 0.2327      | 0.2296      | 0.2266      | 0.2236      | 0.2206      | 0.2177      | 0.2148      |
| <b>-0.6</b> | 0.2743      | 0.2709      | 0.2676      | 0.2643      | 0.2611      | 0.2578      | 0.2546      | 0.2514      | 0.2483      | 0.2451      |
| <b>-0.5</b> | 0.3085      | 0.3050      | 0.3015      | 0.2981      | 0.2946      | 0.2912      | 0.2877      | 0.2843      | 0.2810      | 0.2776      |
| <b>-0.4</b> | 0.3446      | 0.3409      | 0.3372      | 0.3336      | 0.3300      | 0.3264      | 0.3228      | 0.3192      | 0.3156      | 0.3121      |
| <b>-0.3</b> | 0.3821      | 0.3783      | 0.3745      | 0.3707      | 0.3669      | 0.3632      | 0.3594      | 0.3557      | 0.3520      | 0.3483      |
| <b>-0.2</b> | 0.4207      | 0.4168      | 0.4129      | 0.4090      | 0.4052      | 0.4013      | 0.3974      | 0.3936      | 0.3897      | 0.3859      |
| <b>-0.1</b> | 0.4602      | 0.4562      | 0.4522      | 0.4483      | 0.4443      | 0.4404      | 0.4364      | 0.4325      | 0.4286      | 0.4247      |
| <b>-0.0</b> | 0.5000      | 0.4960      | 0.4920      | 0.4880      | 0.4840      | 0.4801      | 0.4761      | 0.4721      | 0.4681      | 0.4641      |

**Table: Cumulative Probabilities for the Standard Normal Distribution,  $Z \sim \text{Normal}(0,1)$ .**

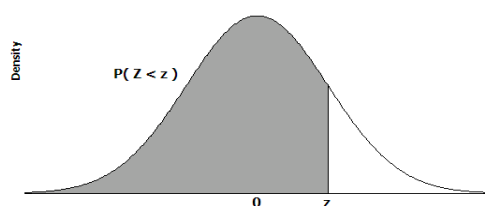
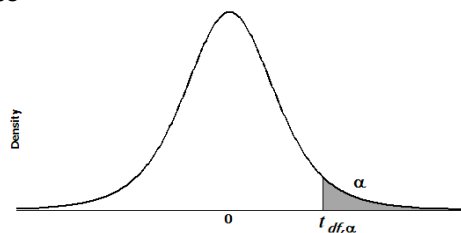
[illegible]

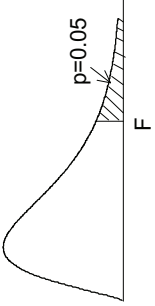
Table: t distribution critical values



| df   | Upper tail probability |       |       |       |       |        |        |        |        |         |         |         |
|------|------------------------|-------|-------|-------|-------|--------|--------|--------|--------|---------|---------|---------|
|      | 0.25                   | 0.20  | 0.15  | 0.10  | 0.05  | 0.025  | 0.02   | 0.01   | 0.005  | 0.0025  | 0.001   | 0.0005  |
| 1    | 1.000                  | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 | 15.895 | 31.821 | 63.657 | 127.321 | 318.309 | 636.619 |
| 2    | 0.816                  | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  | 4.849  | 6.965  | 9.925  | 14.089  | 22.327  | 31.599  |
| 3    | 0.765                  | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  | 3.482  | 4.541  | 5.841  | 7.453   | 10.215  | 12.924  |
| 4    | 0.741                  | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  | 2.999  | 3.747  | 4.604  | 5.598   | 7.173   | 8.610   |
| 5    | 0.727                  | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  | 2.757  | 3.365  | 4.032  | 4.773   | 5.893   | 6.869   |
| 6    | 0.718                  | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  | 2.612  | 3.143  | 3.707  | 4.317   | 5.208   | 5.959   |
| 7    | 0.711                  | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  | 2.517  | 2.998  | 3.499  | 4.029   | 4.785   | 5.408   |
| 8    | 0.706                  | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  | 2.449  | 2.896  | 3.355  | 3.833   | 4.501   | 5.041   |
| 9    | 0.703                  | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  | 2.398  | 2.821  | 3.250  | 3.690   | 4.297   | 4.781   |
| 10   | 0.700                  | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  | 2.359  | 2.764  | 3.169  | 3.581   | 4.144   | 4.587   |
| 11   | 0.697                  | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  | 2.328  | 2.718  | 3.106  | 3.497   | 4.025   | 4.437   |
| 12   | 0.695                  | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  | 2.303  | 2.681  | 3.055  | 3.428   | 3.930   | 4.318   |
| 13   | 0.694                  | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  | 2.282  | 2.650  | 3.012  | 3.372   | 3.852   | 4.221   |
| 14   | 0.692                  | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  | 2.264  | 2.624  | 2.977  | 3.326   | 3.787   | 4.140   |
| 15   | 0.691                  | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  | 2.249  | 2.602  | 2.947  | 3.286   | 3.733   | 4.073   |
| 16   | 0.690                  | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  | 2.235  | 2.583  | 2.921  | 3.252   | 3.686   | 4.015   |
| 17   | 0.689                  | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  | 2.224  | 2.567  | 2.898  | 3.222   | 3.646   | 3.965   |
| 18   | 0.688                  | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  | 2.214  | 2.552  | 2.878  | 3.197   | 3.610   | 3.922   |
| 19   | 0.688                  | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  | 2.205  | 2.539  | 2.861  | 3.174   | 3.579   | 3.883   |
| 20   | 0.687                  | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  | 2.197  | 2.528  | 2.845  | 3.153   | 3.552   | 3.850   |
| 21   | 0.686                  | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  | 2.189  | 2.518  | 2.831  | 3.135   | 3.527   | 3.819   |
| 22   | 0.686                  | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  | 2.183  | 2.508  | 2.819  | 3.119   | 3.505   | 3.792   |
| 23   | 0.685                  | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  | 2.177  | 2.500  | 2.807  | 3.104   | 3.485   | 3.768   |
| 24   | 0.685                  | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  | 2.172  | 2.492  | 2.797  | 3.091   | 3.467   | 3.745   |
| 25   | 0.684                  | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  | 2.167  | 2.485  | 2.787  | 3.078   | 3.450   | 3.725   |
| 26   | 0.684                  | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  | 2.162  | 2.479  | 2.779  | 3.067   | 3.435   | 3.707   |
| 27   | 0.684                  | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  | 2.158  | 2.473  | 2.771  | 3.057   | 3.421   | 3.690   |
| 28   | 0.683                  | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  | 2.154  | 2.467  | 2.763  | 3.047   | 3.408   | 3.674   |
| 29   | 0.683                  | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  | 2.150  | 2.462  | 2.756  | 3.038   | 3.396   | 3.659   |
| 30   | 0.683                  | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  | 2.147  | 2.457  | 2.750  | 3.030   | 3.385   | 3.646   |
| 40   | 0.681                  | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  | 2.123  | 2.423  | 2.704  | 2.971   | 3.307   | 3.551   |
| 50   | 0.679                  | 0.849 | 1.047 | 1.299 | 1.676 | 2.009  | 2.109  | 2.403  | 2.678  | 2.937   | 3.261   | 3.496   |
| 60   | 0.679                  | 0.848 | 1.045 | 1.296 | 1.671 | 2.000  | 2.099  | 2.390  | 2.660  | 2.915   | 3.232   | 3.460   |
| 80   | 0.678                  | 0.846 | 1.043 | 1.292 | 1.664 | 1.990  | 2.088  | 2.374  | 2.639  | 2.887   | 3.195   | 3.416   |
| 100  | 0.677                  | 0.845 | 1.042 | 1.290 | 1.660 | 1.984  | 2.081  | 2.364  | 2.626  | 2.871   | 3.174   | 3.390   |
| 1000 | 0.675                  | 0.842 | 1.037 | 1.282 | 1.646 | 1.962  | 2.056  | 2.330  | 2.581  | 2.813   | 3.098   | 3.300   |
| Z*   | 0.674                  | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  | 2.054  | 2.326  | 2.576  | 2.807   | 3.090   | 3.291   |

# **F distribution for $p = 0.05$**

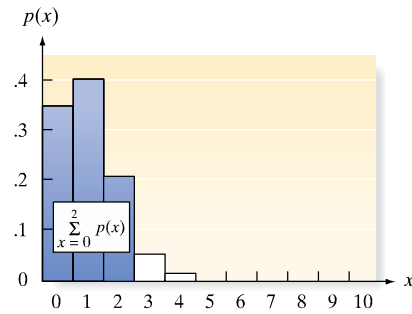
For various pairs of degrees of freedom ( $v_1, v_2$ ), the tabled entries represent the critical values of F above which the proportion  $p = 0.05$  of the distribution falls. (Example, for  $df = 3, 8$  and  $F = 2.92$  is exceeded by  $p = 0.05$  of the population.



| $p = 0.05$ |         | Degrees of freedom for the numerator $v_1$ |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |     |          |
|------------|---------|--------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|-----|----------|
|            |         | $v_2$                                      | 1       | 2       | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 12      | 15      | 20      | 30      | 40      | 60      | 120 | $\infty$ |
| 1          | 161.446 | 199.499                                    | 215.707 | 224.583 | 230.160 | 233.988 | 236.767 | 238.884 | 240.543 | 241.882 | 243.905 | 245.949 | 248.016 | 250.096 | 251.144 | 252.196 | 253.254 | 254.317 |     |          |
| 2          | 18.513  | 19.000                                     | 19.164  | 19.247  | 19.296  | 19.329  | 19.353  | 19.371  | 19.385  | 19.396  | 19.412  | 19.429  | 19.446  | 19.463  | 19.471  | 19.479  | 19.487  | 19.496  |     |          |
| 3          | 10.128  | 9.552                                      | 9.277   | 9.117   | 9.013   | 8.941   | 8.887   | 8.845   | 8.812   | 8.785   | 8.745   | 8.703   | 8.660   | 8.617   | 8.594   | 8.572   | 8.549   | 8.526   |     |          |
| 4          | 7.709   | 6.944                                      | 6.591   | 6.388   | 6.256   | 6.163   | 6.094   | 6.041   | 5.999   | 5.964   | 5.912   | 5.858   | 5.803   | 5.746   | 5.717   | 5.688   | 5.658   | 5.628   |     |          |
| 5          | 6.608   | 5.786                                      | 5.409   | 5.192   | 5.050   | 4.950   | 4.876   | 4.818   | 4.772   | 4.735   | 4.678   | 4.619   | 4.558   | 4.496   | 4.464   | 4.431   | 4.398   | 4.365   |     |          |
| 6          | 5.987   | 5.143                                      | 4.757   | 4.534   | 4.387   | 4.284   | 4.207   | 4.147   | 4.099   | 4.060   | 4.000   | 3.938   | 3.874   | 3.808   | 3.774   | 3.740   | 3.705   | 3.669   |     |          |
| 7          | 5.591   | 4.737                                      | 4.347   | 4.120   | 3.972   | 3.866   | 3.787   | 3.726   | 3.677   | 3.637   | 3.575   | 3.511   | 3.445   | 3.376   | 3.340   | 3.304   | 3.267   | 3.230   |     |          |
| 8          | 5.318   | 4.459                                      | 4.066   | 3.838   | 3.688   | 3.581   | 3.500   | 3.438   | 3.388   | 3.347   | 3.284   | 3.218   | 3.150   | 3.079   | 3.043   | 3.005   | 2.967   | 2.928   |     |          |
| 9          | 5.117   | 4.256                                      | 3.863   | 3.633   | 3.482   | 3.374   | 3.293   | 3.230   | 3.179   | 3.137   | 3.073   | 3.006   | 2.936   | 2.864   | 2.826   | 2.787   | 2.748   | 2.707   |     |          |
| 10         | 4.965   | 4.103                                      | 3.708   | 3.478   | 3.326   | 3.217   | 3.135   | 3.072   | 3.020   | 2.978   | 2.913   | 2.845   | 2.774   | 2.700   | 2.661   | 2.621   | 2.580   | 2.538   |     |          |
| 11         | 4.844   | 3.982                                      | 3.587   | 3.357   | 3.204   | 3.095   | 3.012   | 2.948   | 2.896   | 2.854   | 2.788   | 2.719   | 2.646   | 2.570   | 2.531   | 2.490   | 2.448   | 2.404   |     |          |
| 12         | 4.747   | 3.885                                      | 3.490   | 3.259   | 3.106   | 2.996   | 2.913   | 2.849   | 2.796   | 2.753   | 2.687   | 2.617   | 2.544   | 2.466   | 2.426   | 2.384   | 2.341   | 2.296   |     |          |
| 13         | 4.667   | 3.806                                      | 3.411   | 3.179   | 3.025   | 2.915   | 2.832   | 2.767   | 2.714   | 2.671   | 2.604   | 2.533   | 2.459   | 2.380   | 2.339   | 2.297   | 2.252   | 2.206   |     |          |
| 14         | 4.600   | 3.739                                      | 3.344   | 3.112   | 2.958   | 2.848   | 2.764   | 2.699   | 2.646   | 2.602   | 2.534   | 2.463   | 2.388   | 2.308   | 2.266   | 2.223   | 2.178   | 2.131   |     |          |
| 15         | 4.543   | 3.682                                      | 3.287   | 3.056   | 2.901   | 2.790   | 2.707   | 2.641   | 2.588   | 2.544   | 2.475   | 2.403   | 2.328   | 2.247   | 2.204   | 2.160   | 2.114   | 2.066   |     |          |
| 16         | 4.494   | 3.634                                      | 3.239   | 3.007   | 2.852   | 2.741   | 2.657   | 2.591   | 2.538   | 2.494   | 2.425   | 2.352   | 2.276   | 2.194   | 2.151   | 2.106   | 2.059   | 2.010   |     |          |
| 17         | 4.451   | 3.592                                      | 3.197   | 2.965   | 2.810   | 2.699   | 2.614   | 2.548   | 2.494   | 2.450   | 2.381   | 2.308   | 2.230   | 2.148   | 2.104   | 2.058   | 2.011   | 1.960   |     |          |
| 18         | 4.414   | 3.555                                      | 3.160   | 2.928   | 2.773   | 2.661   | 2.577   | 2.510   | 2.456   | 2.412   | 2.342   | 2.269   | 2.191   | 2.107   | 2.063   | 2.017   | 1.968   | 1.917   |     |          |
| 19         | 4.381   | 3.522                                      | 3.127   | 2.895   | 2.740   | 2.628   | 2.544   | 2.477   | 2.423   | 2.378   | 2.308   | 2.234   | 2.155   | 2.071   | 2.026   | 1.980   | 1.930   | 1.878   |     |          |
| 20         | 4.351   | 3.493                                      | 3.098   | 2.866   | 2.711   | 2.599   | 2.514   | 2.447   | 2.393   | 2.348   | 2.278   | 2.203   | 2.124   | 2.039   | 1.994   | 1.946   | 1.896   | 1.843   |     |          |
| 21         | 4.325   | 3.467                                      | 3.072   | 2.840   | 2.685   | 2.573   | 2.488   | 2.420   | 2.366   | 2.321   | 2.250   | 2.176   | 2.096   | 2.010   | 1.965   | 1.916   | 1.866   | 1.812   |     |          |
| 22         | 4.301   | 3.443                                      | 3.049   | 2.817   | 2.661   | 2.549   | 2.464   | 2.397   | 2.342   | 2.297   | 2.226   | 2.151   | 2.071   | 1.984   | 1.938   | 1.889   | 1.838   | 1.783   |     |          |
| 23         | 4.279   | 3.422                                      | 3.028   | 2.796   | 2.640   | 2.528   | 2.442   | 2.375   | 2.320   | 2.275   | 2.204   | 2.128   | 2.048   | 1.961   | 1.914   | 1.865   | 1.813   | 1.757   |     |          |
| 24         | 4.260   | 3.403                                      | 3.009   | 2.776   | 2.621   | 2.508   | 2.423   | 2.355   | 2.300   | 2.255   | 2.183   | 2.108   | 2.027   | 1.939   | 1.892   | 1.842   | 1.790   | 1.733   |     |          |
| 30         | 4.171   | 3.316                                      | 2.922   | 2.690   | 2.534   | 2.421   | 2.334   | 2.266   | 2.211   | 2.165   | 2.092   | 2.015   | 1.932   | 1.841   | 1.792   | 1.740   | 1.683   | 1.622   |     |          |
| 40         | 4.085   | 3.232                                      | 2.839   | 2.606   | 2.449   | 2.336   | 2.249   | 2.180   | 2.124   | 2.077   | 2.003   | 1.924   | 1.839   | 1.744   | 1.693   | 1.637   | 1.577   | 1.509   |     |          |
| 60         | 4.001   | 3.150                                      | 2.758   | 2.525   | 2.368   | 2.254   | 2.167   | 2.097   | 2.040   | 1.993   | 1.917   | 1.836   | 1.748   | 1.649   | 1.594   | 1.534   | 1.467   | 1.389   |     |          |
| 120        | 3.920   | 3.072                                      | 2.680   | 2.447   | 2.290   | 2.175   | 2.087   | 2.016   | 1.959   | 1.910   | 1.834   | 1.750   | 1.659   | 1.554   | 1.495   | 1.429   | 1.352   | 1.254   |     |          |
| $\infty$   | 3.841   | 2.996                                      | 2.605   | 2.372   | 2.214   | 2.099   | 2.010   | 1.938   | 1.880   | 1.831   | 1.752   | 1.666   | 1.571   | 1.459   | 1.394   | 1.318   | 1.221   | 1.000   |     |          |



TABLE II Binomial Probabilities



Tabulated values are  $\sum_{x=0}^k p(x)$ . (Computations are rounded at the third decimal place.)

**a.  $n = 5$**

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30  | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|
| 0                                    | .951  | .774  | .590  | .328  | .168 | .078 | .031 | .010 | .002 | .000 | .000 | .000 | .000 |
| 1                                    | .999  | .977  | .919  | .737  | .528 | .337 | .188 | .087 | .031 | .007 | .000 | .000 | .000 |
| 2                                    | 1.000 | .999  | .991  | .942  | .837 | .683 | .500 | .317 | .163 | .058 | .009 | .001 | .000 |
| 3                                    | 1.000 | 1.000 | 1.000 | .993  | .969 | .913 | .812 | .663 | .472 | .263 | .081 | .023 | .001 |
| 4                                    | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .990 | .969 | .922 | .832 | .672 | .410 | .226 | .049 |

**b.  $n = 6$**

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30  | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|
| 0                                    | .941  | .735  | .531  | .262  | .118 | .047 | .016 | .004 | .001 | .000 | .000 | .000 | .000 |
| 1                                    | .999  | .967  | .886  | .655  | .420 | .233 | .109 | .041 | .011 | .002 | .000 | .000 | .000 |
| 2                                    | 1.000 | .998  | .984  | .901  | .744 | .544 | .344 | .179 | .070 | .017 | .001 | .000 | .000 |
| 3                                    | 1.000 | 1.000 | .999  | .983  | .930 | .821 | .656 | .456 | .256 | .099 | .016 | .002 | .000 |
| 4                                    | 1.000 | 1.000 | 1.000 | .998  | .989 | .959 | .891 | .767 | .580 | .345 | .114 | .033 | .001 |
| 5                                    | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .996 | .984 | .953 | .882 | .738 | .469 | .265 | .059 |

**c.  $n = 7$**

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30   | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|
| 0                                    | .932  | .698  | .478  | .210  | .082  | .028 | .008 | .002 | .000 | .000 | .000 | .000 | .000 |
| 1                                    | .998  | .956  | .850  | .577  | .329  | .159 | .063 | .019 | .004 | .000 | .000 | .000 | .000 |
| 2                                    | 1.000 | .996  | .974  | .852  | .647  | .420 | .227 | .096 | .029 | .005 | .000 | .000 | .000 |
| 3                                    | 1.000 | 1.000 | .997  | .967  | .874  | .710 | .500 | .290 | .126 | .033 | .003 | .000 | .000 |
| 4                                    | 1.000 | 1.000 | 1.000 | .995  | .971  | .904 | .773 | .580 | .353 | .148 | .026 | .004 | .000 |
| 5                                    | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .981 | .937 | .841 | .671 | .423 | .150 | .044 | .002 |
| 6                                    | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .992 | .972 | .918 | .790 | .522 | .302 | .068 |

(continued)

TABLE II (continued)

d.  $n = 8$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|
| 0                                              | .923  | .663  | .430  | .168  | .058  | .017 | .004 | .001 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .997  | .943  | .813  | .503  | .255  | .106 | .035 | .009 | .001 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .994  | .962  | .797  | .552  | .315 | .145 | .050 | .011 | .001 | .000 | .000 | .000 |
| 3                                              | 1.000 | 1.000 | .995  | .944  | .806  | .594 | .363 | .174 | .058 | .010 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | 1.000 | .990  | .942  | .826 | .637 | .406 | .194 | .056 | .005 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .999  | .989  | .950 | .855 | .685 | .448 | .203 | .038 | .006 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .991 | .965 | .894 | .745 | .497 | .187 | .057 | .003 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .996 | .983 | .942 | .832 | .570 | .337 | .077 |

e.  $n = 9$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|
| 0                                              | .914  | .630  | .387  | .134  | .040  | .010  | .002 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .997  | .929  | .775  | .436  | .196  | .071  | .020 | .004 | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .992  | .947  | .738  | .463  | .232  | .090 | .025 | .004 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .999  | .992  | .914  | .730  | .483  | .254 | .099 | .025 | .003 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | .999  | .980  | .901  | .733  | .500 | .267 | .099 | .020 | .001 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .997  | .975  | .901  | .746 | .517 | .270 | .086 | .008 | .001 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .975  | .910 | .768 | .537 | .262 | .053 | .008 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .980 | .929 | .804 | .564 | .225 | .071 | .003 |
| 8                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .990 | .960 | .866 | .613 | .370 | .086 |

f.  $n = 10$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|
| 0                                              | .904  | .599  | .349  | .107  | .028  | .006  | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .996  | .914  | .736  | .376  | .149  | .046  | .011 | .002 | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .988  | .930  | .678  | .383  | .167  | .055 | .012 | .002 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .999  | .987  | .879  | .650  | .382  | .172 | .055 | .011 | .001 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | .998  | .967  | .850  | .633  | .377 | .166 | .047 | .006 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .994  | .953  | .834  | .623 | .367 | .150 | .033 | .002 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | .999  | .989  | .945  | .828 | .618 | .350 | .121 | .013 | .001 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .988  | .945 | .833 | .617 | .322 | .070 | .012 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .989 | .954 | .851 | .624 | .264 | .086 | .004 |
| 9                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .994 | .972 | .893 | .651 | .401 | .096 |

(continued)

TABLE II (continued)

g.  $n = 15$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50   | .60   | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|
| 0                                              | .860  | .463  | .206  | .035  | .005  | .000  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .990  | .829  | .549  | .167  | .035  | .005  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .964  | .816  | .398  | .127  | .027  | .004  | .000  | .000 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .995  | .944  | .648  | .297  | .091  | .018  | .002  | .000 | .000 | .000 | .000 | .000 |
| 4                                              | 1.000 | .999  | .987  | .838  | .515  | .217  | .059  | .009  | .001 | .000 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | .998  | .939  | .722  | .403  | .151  | .034  | .004 | .000 | .000 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | .982  | .869  | .610  | .304  | .095  | .015 | .001 | .000 | .000 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | .996  | .950  | .787  | .500  | .213  | .050 | .004 | .000 | .000 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | .999  | .985  | .905  | .696  | .390  | .131 | .018 | .000 | .000 | .000 |
| 9                                              | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .966  | .849  | .597  | .278 | .061 | .002 | .000 | .000 |
| 10                                             | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .991  | .941  | .783  | .485 | .164 | .013 | .001 | .000 |
| 11                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .982  | .909  | .703 | .352 | .056 | .005 | .000 |
| 12                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .973  | .873 | .602 | .184 | .036 | .000 |
| 13                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .995  | .965 | .833 | .451 | .171 | .010 |
| 14                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .995 | .965 | .794 | .537 | .140 |

h.  $n = 20$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50   | .60   | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|
| 0                                              | .818  | .358  | .122  | .012  | .001  | .000  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .983  | .736  | .392  | .069  | .008  | .001  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 2                                              | .999  | .925  | .677  | .206  | .035  | .004  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .984  | .867  | .411  | .107  | .016  | .001  | .000  | .000 | .000 | .000 | .000 | .000 |
| 4                                              | 1.000 | .997  | .957  | .630  | .238  | .051  | .006  | .000  | .000 | .000 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | .989  | .804  | .416  | .126  | .021  | .002  | .000 | .000 | .000 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | .998  | .913  | .608  | .250  | .058  | .006  | .000 | .000 | .000 | .000 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | .968  | .772  | .416  | .132  | .021  | .001 | .000 | .000 | .000 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | .990  | .887  | .596  | .252  | .057  | .005 | .000 | .000 | .000 | .000 |
| 9                                              | 1.000 | 1.000 | 1.000 | .997  | .952  | .755  | .412  | .128  | .017 | .001 | .000 | .000 | .000 |
| 10                                             | 1.000 | 1.000 | 1.000 | .999  | .983  | .872  | .588  | .245  | .048 | .003 | .000 | .000 | .000 |
| 11                                             | 1.000 | 1.000 | 1.000 | 1.000 | .995  | .943  | .748  | .404  | .113 | .010 | .000 | .000 | .000 |
| 12                                             | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .979  | .868  | .584  | .228 | .032 | .000 | .000 | .000 |
| 13                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .994  | .942  | .750  | .392 | .087 | .002 | .000 | .000 |
| 14                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .979  | .874  | .584 | .196 | .011 | .000 | .000 |
| 15                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .994  | .949  | .762 | .370 | .043 | .003 | .000 |
| 16                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .984  | .893 | .589 | .133 | .016 | .000 |
| 17                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .965 | .794 | .323 | .075 | .001 |
| 18                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .992 | .931 | .608 | .264 | .017 |
| 19                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .988 | .878 | .642 | .182 |

(continued)

**TABLE XIV** Critical Values of  $T_L$  and  $T_U$  for the Wilcoxon Rank Sum Test: Independent Samples

*Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).*

**a.  $\alpha = .025$  one-tailed;  $\alpha = .05$  two-tailed**

| $n_2 \backslash n_1$ | 3     |       | 4     |       | 5     |       | 6     |       | 7     |       | 8     |       | 9     |       | 10    |       |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                      | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ |
| <b>3</b>             | 5     | 16    | 6     | 18    | 6     | 21    | 7     | 23    | 7     | 26    | 8     | 28    | 8     | 31    | 9     | 33    |
| <b>4</b>             | 6     | 18    | 11    | 25    | 12    | 28    | 12    | 32    | 13    | 35    | 14    | 38    | 15    | 41    | 16    | 44    |
| <b>5</b>             | 6     | 21    | 12    | 28    | 18    | 37    | 19    | 41    | 20    | 45    | 21    | 49    | 22    | 53    | 24    | 56    |
| <b>6</b>             | 7     | 23    | 12    | 32    | 19    | 41    | 26    | 52    | 28    | 56    | 29    | 61    | 31    | 65    | 32    | 70    |
| <b>7</b>             | 7     | 26    | 13    | 35    | 20    | 45    | 28    | 56    | 37    | 68    | 39    | 73    | 41    | 78    | 43    | 83    |
| <b>8</b>             | 8     | 28    | 14    | 38    | 21    | 49    | 29    | 61    | 39    | 73    | 49    | 87    | 51    | 93    | 54    | 98    |
| <b>9</b>             | 8     | 31    | 15    | 41    | 22    | 53    | 31    | 65    | 41    | 78    | 51    | 93    | 63    | 108   | 66    | 114   |
| <b>10</b>            | 9     | 33    | 16    | 44    | 24    | 56    | 32    | 70    | 43    | 83    | 54    | 98    | 66    | 114   | 79    | 131   |

**b.  $\alpha = .05$  one-tailed;  $\alpha = .10$  two-tailed**

| $n_2 \backslash n_1$ | 3     |       | 4     |       | 5     |       | 6     |       | 7     |       | 8     |       | 9     |       | 10    |       |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                      | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ |
| <b>3</b>             | 6     | 15    | 7     | 17    | 7     | 20    | 8     | 22    | 9     | 24    | 9     | 27    | 10    | 29    | 11    | 31    |
| <b>4</b>             | 7     | 17    | 12    | 24    | 13    | 27    | 14    | 30    | 15    | 33    | 16    | 36    | 17    | 39    | 18    | 42    |
| <b>5</b>             | 7     | 20    | 13    | 27    | 19    | 36    | 20    | 40    | 22    | 43    | 24    | 46    | 25    | 50    | 26    | 54    |
| <b>6</b>             | 8     | 22    | 14    | 30    | 20    | 40    | 28    | 50    | 30    | 54    | 32    | 58    | 33    | 63    | 35    | 67    |
| <b>7</b>             | 9     | 24    | 15    | 33    | 22    | 43    | 30    | 54    | 39    | 66    | 41    | 71    | 43    | 76    | 46    | 80    |
| <b>8</b>             | 9     | 27    | 16    | 36    | 24    | 46    | 32    | 58    | 41    | 71    | 52    | 84    | 54    | 90    | 57    | 95    |
| <b>9</b>             | 10    | 29    | 17    | 39    | 25    | 50    | 33    | 63    | 43    | 76    | 54    | 90    | 66    | 105   | 69    | 111   |
| <b>10</b>            | 11    | 31    | 18    | 42    | 26    | 54    | 35    | 67    | 46    | 80    | 57    | 95    | 69    | 111   | 83    | 127   |

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.